Definition 12.5: Meet of A and B

$$A \wedge B = [a_{ij}] \wedge [b_{ij}] = [a_{ij} \wedge b_{ij}] = [c_{ij}] \text{ where } c_{ij} = \begin{cases} 1, & \text{if both } a_{ij} = 1 \text{ and } b_{ij} = 1 \\ 0, & \text{if either } a_{ij} = 0 \text{ or } b_{ij} = 0. \end{cases}$$

It is clear that $(a \vee b) = \max\{a, b\}$; $(a \wedge b) = \min\{a, b\}$, $a, b \in \{0, 1\}$.

Example 12.8

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ be any two boolean matrices of the same type. Find $A \vee B$ and $A \wedge B$.

Solution

$$\text{Then } A \vee B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 \vee 1 & 1 \vee 1 \\ 1 \vee 0 & 1 \vee 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A \wedge B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \wedge \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 \wedge 1 & 1 \wedge 1 \\ 1 \wedge 0 & 1 \wedge 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

Properties satisfied by join and meet

Let $\mathbb{B}$ be the set of all boolean matrices of the same type. We only state the properties of meet and join.

Closure property

$A, B \in \mathbb{B}$, $A \vee B = [a_{ij}] \vee [b_{ij}] = [a_{ij} \vee b_{ij}] \in \mathbb{B}$. (Because, $(a_{ij} \vee b_{ij})$ is either 0 or 1 $\forall i, j$. $\vee$ is a binary operation on $\mathbb{B}$.

Associative property

$A \vee (B \vee C) = (A \vee B) \vee C$, $\forall A, B, C \in \mathbb{B}$. $\vee$ is associative.

Existence of identity property

$\forall A \in \mathbb{B}$, $\exists$ the null matrix $0 \in \mathbb{B} \ni A \vee 0 = 0 \vee A = A$. The identity element for $\vee$ is the null matrix.

Existence of inverse property

For any matrix $A \in \mathbb{B}$, it is impossible to find a matrix

$B \in \mathbb{B} \ni A \vee B = B \vee A = 0$. So the inverse does not exist.

Similarly, it can be verified that the operation meet $\wedge$ satisfies (i) closure property (ii) commutative property (iii) associative property (iv) the matrix $U = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ exists as the identity in $\mathbb{B}$ and (v) the existence of inverse is not assured.

12.2.4 Modular Arithmetic

Having discussed the properties of operations like basic usual arithmetic operations, matrix addition and multiplication, join and meet of boolean matrices, one more new operation called the **Modular Arithmetic** is discussed in this section**.** The modular arithmetic refers to the process of **dividing** some number a by a positive integer n (> 1), called modulus, and then equating a with the remainder b modulo n and it is written as $a \equiv b \pmod{n}$, read as 'a is congruent to b modulo n'.

12th_Maths_Vol 2_EM_CH 12_Discrete Mathematics.indd 232 16-12-2022 16:54:19